

MOHAMED KHALID M JAFFAR

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EDUCATION

University of Maryland, College Park, USA
Doctor of Philosophy (PhD) in Aerospace Engineering

Expected Fall '25
GPA: 3.62/4

Indian Institute of Technology Madras, India
Dual Degree (B.Tech & M.Tech) in Aerospace Engineering
Minor in Robotics

2013 – 2018
GPA: 9.06/10
Class Rank 1

AWARDS AND HONORS

- Ann G. Wylie **Dissertation Fellowship** for pursuing *semester-long* academic research (Fall 2024)
- **Gold Medalist** for **best Academic record** in Dual Degree Aerospace Engineering (2018)
- Recipient of IIT Madras' **IC&SR Innovative Project grant** (2017)
- Prof. E. G. Tulapurkara **Institute Merit Prize** for best academic performance (2015-16)
- **NTU-India connect Scholarship** from NTU, Singapore; *1 among 20 students from India* (2016)
- Within *top 30 ranks in State* (Tamil Nadu) in **Regional Mathematics Olympiad (RMO)** (2010)

SKILLS AND EXPERTISE

- **Conceptual Expertise:** Robot Autonomy, Applied Mathematics, Motion Planning and Controls, Graph Theory, Neural Networks and Large Language Models (LLMs)
- **Math background:** Linear Algebra, Numerical Optimization, Probability and Stochastic Processes, Differential Geometry, Data structures and Algorithms
- **Programming Languages:** Python, Object-Oriented MATLAB, C++
- **Tools and Softwares:** ROS, Simulink, System Identification Toolbox, SolidWorks, PyTorch

RESEARCH EXPERIENCE

Safe feedback motion re-planning of robots in dynamic spaces (*Doctoral Thesis*)
Advisor: Dr. Michael Otte, Motion and Teaming Lab, University of Maryland

- Design and testing of motion planning algorithms that respect robots' controllability constraints
- Propose and validate failsafe algorithms for robot's safe operation despite drastic physical changes to the system
- Computing optimal paths with guaranteed safety for robots in environments with moving obstacles
- **Book chapter:** "PiP-X: Funnel-based Online Feedback Motion Planning/Replanning in Dynamic Environments." *Algorithmic Foundations of Robotics (2022)* [*best paper award consideration*]

Long-range navigation of legged robots in unmapped outdoor terrain **Summer '24**
Advisor: Jason Pusey, U.S. Army Research Lab & Dr. Dinesh Manocha, University of Maryland

- Field experiments of quadruped autonomous navigation in dense forested spaces, by developing algorithms consisting of graph-based planner to route, and a local re-planner to maneuver
- Robot-gait adaptation to tread uneven terrains, by leveraging visual perception using vision language models (VLM), and haptic perception using inertial sensors
- Multi-layer terrain mapping using intensity data from LiDAR sensor
- **Article preprint:** "Robot Navigation Using Physically Grounded Vision-Language Models in Outdoor Environments." to appear at *International Conference of Robotics & Automation (2025)*

Maneuver control of flying robots in indoor spaces

(*Masters' Thesis*)

Advisor: Dr. Ranjith Mohan & Dr. Asokan Thondiyath, IIT Madras

- On-board autonomy for aggressive maneuver control of quadrotors in tight, indoor spaces
- Synthesized and compared performance of different nonlinear control strategies
- Custom-designed and fabricated an autopilot board consisting of STM microcontrollers and inertial sensors for stabilisation
- **Paper:** "A Novel Guidance Algorithm and Comparison of Nonlinear Control Strategies Applied to an Indoor Quadrotor." *Fifth Indian Control Conference* (2019)

Autonomous target-tracking unmanned aerial vehicle (UAV)

Jun – Nov 2015

Collaboration project between Defense R&D Organisation (DRDO) & IIT Madras

- Python programming of a UAV to track and follow a ground target autonomously
- Modeled the system in MATLAB, implementing a robust mathematical model of the system and an efficient PID controller to make it follow an optimal trajectory

STARTUP EXPERIENCE

Implementation of autonomous control for a flying robot

(2015–17)

Lead Product Developer & **Founding member**, *DeTect Technologies*, an energy-sector startup

- Developed algorithms and programmed the quadcopter using Python for autonomously tracing a predefined route with precision, to inspect oil and gas pipelines
- Carried out field tests for System Identification and evaluation of different controller strategies

APPLIED PROJECT EXPERIENCE

Development of an Airbrake system to control a Rocket's ascent

(2021–24)

Terrapin Rocket team for Spaceport America Cup in Las Cruces, New Mexico

Awards: Placed **category 2nd** in 2023 and **overall 1st** in 2024, out of **150 teams globally**

- Design and fabrication of electro-mechanical modules of outward-deploying braking flaps
- Synthesized an autonomy block for the rocket to reach a target altitude, involving programming in C++ and assembly-level language
- **Technical Article:** "The Development of a Drag-Modulating Feedback System to Control a Rocket's Ascent." *AIAA SciTech Conference* (2024)

Design and fabrication of radio-controlled (RC) fixed-wing airplane

(2016–17)

IIT Madras team for SAE Aero Design student competition 2017 in Fortworth, Texas

Recognition: Placed **12th** in world and **4th** in India, out of **75 teams globally**

- Carried out analysis of aerodynamic characteristics, stability and performance of the RC aircraft
- Designed the aircraft structure using SolidWorks and manufactured using composites

PROFESSIONAL SERVICE AND LEADERSHIP

- Mentoring and **Supervising undergraduate students** at Motion and Teaming Lab, UMD in robotics-related projects (2022–24)
- **Academic and Professional Chair** at Graduate Student Advisory Committee (GSAC) of Aerospace Department, UMD College Park (2021–22)
- **Student representative in Executive Committee** of Maryland Robotics Center (2020–21)
- **Lab Instructor** for course: *Flight Controls and Simulation*. Conducted lab sessions, lectures and evaluated reports, final papers (2017–18)
- **Project Mentor** for student-teams in **Robotics & Aero Clubs** at the Centre for Innovation (CFI), IIT Madras (2016–18)